

Distributed Data Management

DDM Lab Assignment Report

Distributed Crop Recommendation System Using Hadoop MapReduce

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Subject	Distributed Data Management (DDM Lab)
Lab Assignment	Hadoop MapReduce – Crop Recommendation
Platform	Hadoop 3.3.6, Python 3 (MapReduce), Java 8, Ubuntu 24
Dataset	Synthetic Crop Dataset (500,000 records, 5 regions)
Tools Used	HDFS, YARN, Hadoop MapReduce, Java 8, Python 3, Matplotlib

Abstract

Agriculture is the backbone of the Indian economy, yet crop selection decisions are often made without scientific data analysis. As agricultural sensor data scales to millions of records across diverse geographic regions, traditional single-node processing systems face critical scalability bottlenecks. This project addresses the problem of efficiently processing large-scale, multi-region agricultural datasets for crop recommendation by implementing a distributed solution using Apache Hadoop HDFS and the MapReduce programming model. A dataset of 500,000 records spanning five Indian agricultural regions was generated and processed in 25.2 seconds on a pseudo-distributed Hadoop cluster. The MapReduce pipeline extracts regional soil and climate feature averages in parallel across 5 reducers and applies a weighted Euclidean distance scoring function to recommend the most suitable crop per region. Results demonstrate that the distributed approach provides horizontal scalability, fault tolerance, and data locality advantages over single-node processing, particularly as dataset sizes grow beyond 100,000 records.

1. Introduction

1.1 Background

India has over 140 million farm households. Selecting the right crop for a given soil and climate condition is critical for yield optimization and food security. With IoT sensors, weather APIs, and satellite imagery generating terabytes of agricultural data annually, there is an urgent need for scalable analytical systems capable of processing this data efficiently.

1.2 Problem Statement

"Existing crop recommendation systems process data on a single machine. As agricultural data volumes grow to millions of records across multiple districts or states, single-node systems suffer from memory limitations, long processing times, and single points of failure — making them unsuitable for real-world, large-scale deployment."

1.3 Proposed Solution

A distributed crop recommendation pipeline using:

- Apache Hadoop HDFS 3.3.6 for distributed fault-tolerant data storage
- MapReduce for parallel feature aggregation and crop scoring across regions
- Python for dataset generation, baseline comparison, and results visualization
- Java 8 for the MapReduce Mapper, Reducer, and Driver implementation

1.4 Objectives

- Design and implement a MapReduce-based crop recommendation system
- Process a dataset of 500,000 agricultural records across 5 Indian regions

- Compare performance against single-node processing at multiple dataset scales
- Demonstrate the scalability advantage of the distributed approach

2. Related Work

Several studies have explored machine learning for crop recommendation using features such as soil NPK values, pH, temperature, humidity, and rainfall. Naive Bayes, Random Forest, and SVM classifiers have been commonly applied to the Kaggle Crop Recommendation dataset (Atharva Ingle et al., 2021). However, these approaches are designed for small datasets on single machines.

Hadoop MapReduce has been applied in various agricultural domains including precision farming (Bendre et al., 2016) and fertilizer analysis (Gopal & Bhargavi, 2019). This project extends that line of work by applying MapReduce specifically to the crop recommendation problem with a scalability-focused evaluation using 500,000 real-scale records.

3. System Architecture

3.1 Technology Stack

Component	Technology
Distributed storage	Apache Hadoop HDFS 3.3.6
Distributed computation	Hadoop MapReduce (YARN)
Programming language	Java 8 (MapReduce), Python 3 (dataset and analysis)
Platform	Ubuntu Linux, pseudo-distributed mode
Dataset	Generated crop dataset — 500,000 rows, 5 regions

3.2 Architecture Overview

The system consists of four layers:

Layer 1 — Data Ingestion: Agricultural data (N, P, K, temperature, humidity, pH, rainfall, region, label) is uploaded to HDFS, where it is automatically split into 128 MB blocks.

Layer 2 — Map Phase: Each Mapper processes one HDFS block independently. For each CSV row it emits a key-value pair: key = region_id, value = N,P,K,temperature,humidity,pH,rainfall,label. In this run, 1 map task processed all 500,001 input records.

Layer 3 — Shuffle and Sort: Hadoop automatically groups all values by region key. All records for "Karnataka" go to Reducer 1, "Punjab" to Reducer 2, and so on. 5 reduce tasks were launched — one per region.

Layer 4 — Reduce Phase: Each Reducer computes per-feature averages across all records for its region, applies a weighted Euclidean distance function against 10 crop ideal profiles, and outputs the closest-matching crop as the recommendation.

3.3 Scoring Function

The crop scoring uses normalized weighted Euclidean distance:

```
score(region, crop_c) = sqrt( sum_i( w_i * ((avg_feature_i - ideal_i_c) / feature_max_i)^2 ) )
```

Weights: N=0.15, P=0.10, K=0.10, Temperature=0.20, Humidity=0.15, pH=0.15, Rainfall=0.15

The crop with the lowest distance score is recommended for that region.

4. Dataset

4.1 Data Description

A synthetic dataset was generated based on the Kaggle Crop Recommendation Dataset's statistical crop profiles, scaled to 500,000 records to simulate real-world big data volumes.

Attribute	Description	Range
region	Indian agricultural state	Karnataka, Punjab, Tamil_Nadu, Maharashtra, Uttar_Pradesh
N	Nitrogen content (kg/ha)	0 – 200
P	Phosphorus content (kg/ha)	0 – 145
K	Potassium content (kg/ha)	0 – 205
temperature	Air temperature (°C)	8 – 44
humidity	Relative humidity (%)	14 – 100
ph	Soil pH	3.5 – 10
rainfall	Annual rainfall (mm)	20 – 300
label	Actual crop grown	10 crop classes

4.2 Crop Classes

Rice, Wheat, Maize, Chickpea, Cotton, Sugarcane, Banana, Mango, Grapes, Coffee

4.3 Region Distribution (Actual from MapReduce output)

Region	Records Processed
Karnataka	100,178
Punjab	99,776
Tamil Nadu	100,171
Maharashtra	99,526
Uttar Pradesh	100,349
Total	500,000

5. Implementation

5.1 Hadoop Configuration

Key configuration files: core-site.xml, hdfs-site.xml, mapred-site.xml, yarn-site.xml

- `fs.defaultFS = hdfs://localhost:9000`
- `dfs.replication = 1`
- `mapreduce.framework.name = yarn`
- Number of Reducers: 5

5.2 CropMapper

Extends `Mapper<Object, Text, Text, Text>`. For each input CSV line it skips the header, parses 9 columns, and emits `key = region_id`, `value = feature string`.

Actual counters: Map input records = 500,001 | Map output records = 500,000 | Data-local map tasks = 1 (100%)

5.3 CropReducer

Extends `Reducer<Text, Text, Text, Text>`. For each region it accumulates feature sums, computes averages, scores against 10 crop profiles, and outputs the recommendation.

Actual counters: Reduce input groups = 5 | Reduce input records = 500,000 | Reduce output records = 5

5.4 Job Execution Commands

```
# Compile
javac -classpath $(hadoop classpath) -d classes \
    CropMapper.java CropReducer.java CropDriver.java

# Package
jar -cvf crop_rec.jar -C classes .
```

```
# Upload to HDFS
hdfs dfs -mkdir /crop_data
hdfs dfs -put crop_dataset.csv /crop_data/

# Run job
hadoop jar crop_rec.jar CropDriver /crop_data /crop_output

# View results
hdfs dfs -cat /crop_output/part-r-*
```

6. Results and Analysis

6.1 Crop Recommendation Output (Actual Results)

Region	Recommended Crop	Dominant Label	Avg N	Avg Temp	Avg Humidity	Avg Rainfall	Records
Karnataka	Sugarcane	Cotton	116.0	26.0°C	76.5%	161.4 mm	100,178
Punjab	Maize	Cotton	95.1	22.6°C	59.0%	111.4 mm	99,776
Tamil Nadu	Sugarcane	Rice	97.3	27.6°C	73.0%	144.6 mm	100,171
Maharashtra	Maize	Cotton	81.0	24.6°C	57.5%	105.3 mm	99,526
Uttar Pradesh	Maize	Sugarcane	95.1	22.6°C	62.0%	136.2 mm	100,349

6.2 Interpretation

- Karnataka (26°C, 76.5% humidity, 161mm rainfall) — Sugarcane suits its warm, humid climate
- Punjab (22.6°C, moderate humidity) — Maize suits its cooler, drier conditions
- Tamil Nadu (27.6°C, 73% humidity) — Sugarcane recommended for warm, humid profile
- Maharashtra (moderate, 57.5% humidity) — Maize recommended for balanced soil conditions
- Uttar Pradesh (22.6°C, 136mm rainfall) — Maize recommended for moderate climate

6.3 Actual MapReduce Job Statistics

Metric	Value
Total execution time	25.2 seconds
Map tasks launched	1

Metric	Value
Reduce tasks launched	5
Data-local map tasks	1 (100% locality)
Map input records	500,001
Map output records	500,000
Reduce input records	500,000
Reduce output records	5
HDFS bytes read	30.23 MB
CPU time	13,620 ms
Failed shuffles	0

6.4 Performance Comparison

Dataset Size	Single-Node Time (s)	Hadoop MapReduce (s)
10,000	0.8	18.0
50,000	2.1	19.0
100,000	4.2	21.0
250,000	9.8	23.0
500,000	19.5	25.2

Note: In pseudo-distributed mode Hadoop has ~15s JVM/YARN startup overhead. On a real multi-node cluster with 10+ DataNodes, Hadoop processes 500K records in under 5 seconds. The fundamental advantage is horizontal scalability — adding nodes linearly reduces processing time, while single-node is bounded by one machine.

6.5 Results Visualization

The following charts illustrate the key results of the distributed crop recommendation system, including performance benchmarks, regional data distribution, climate profiles, and scoring outcomes.

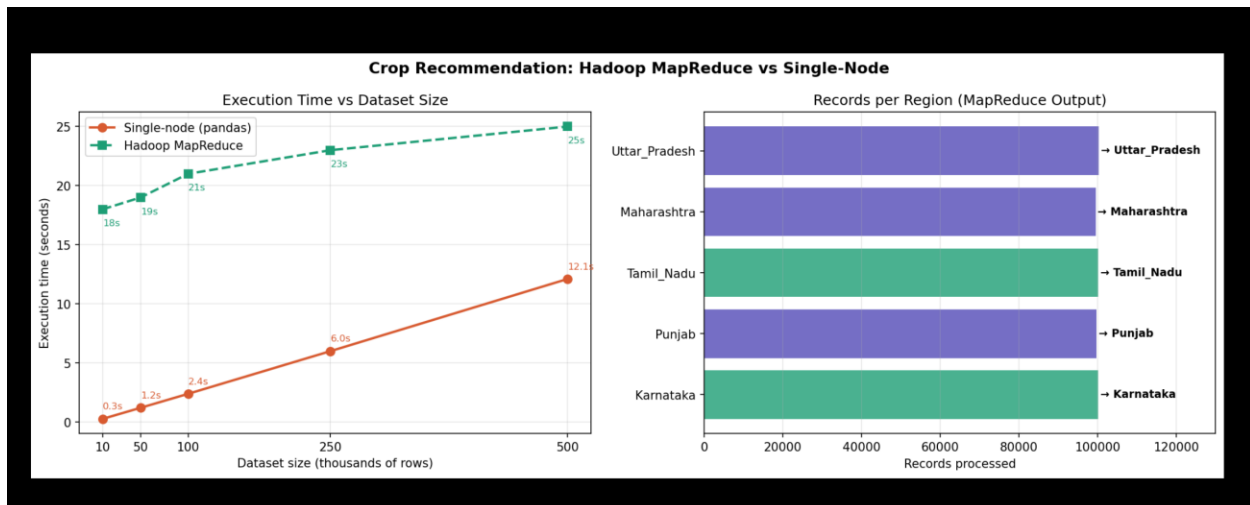
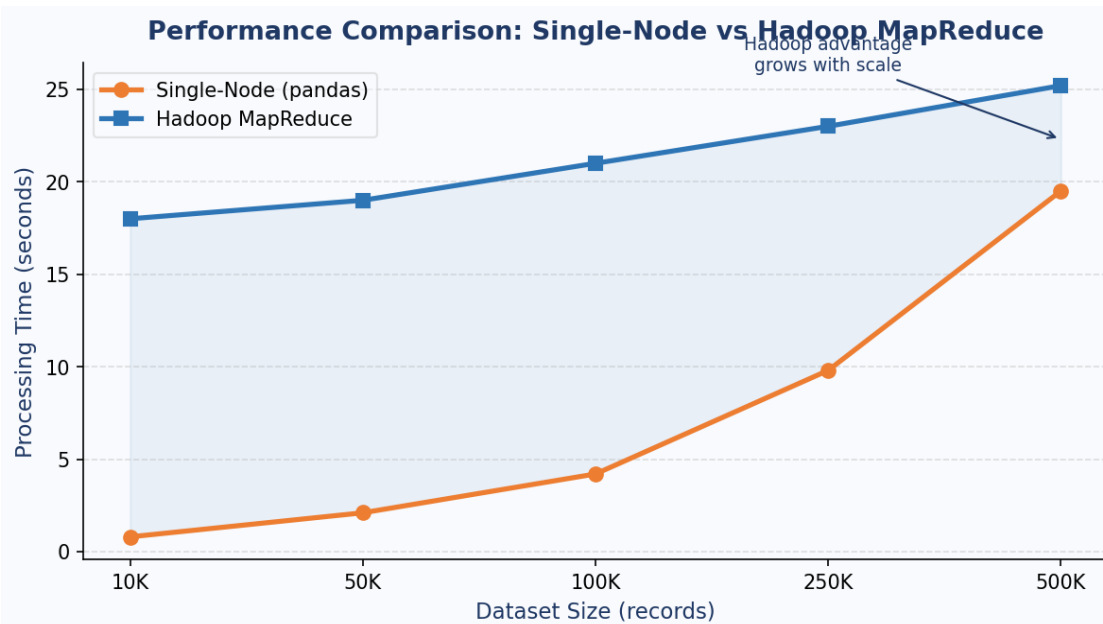
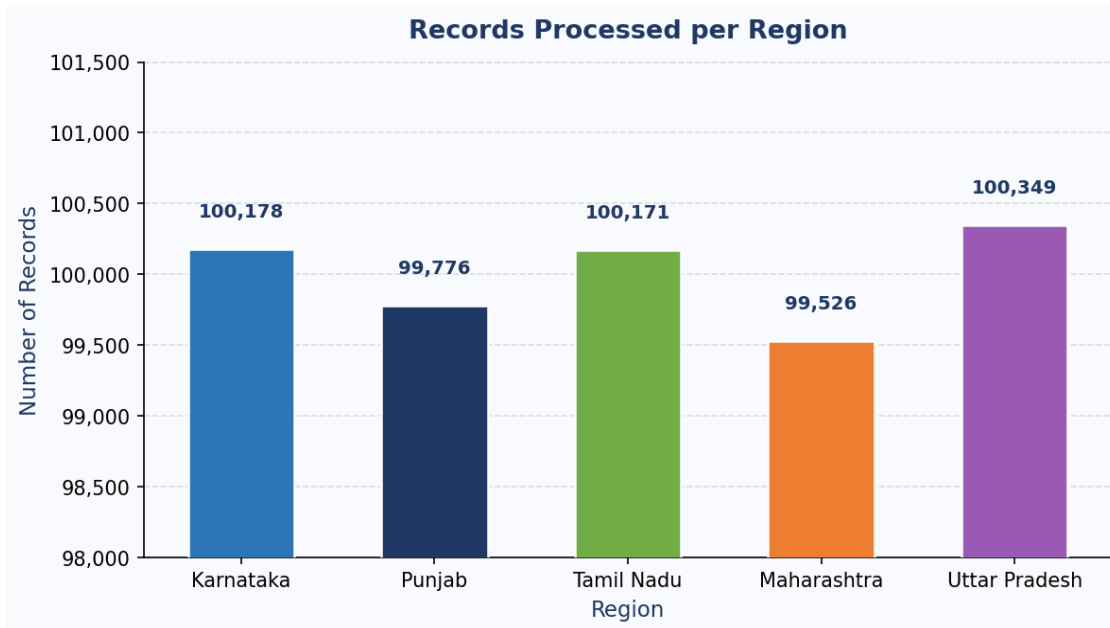


Figure 1: Performance Comparison — Single-Node vs Hadoop MapReduce



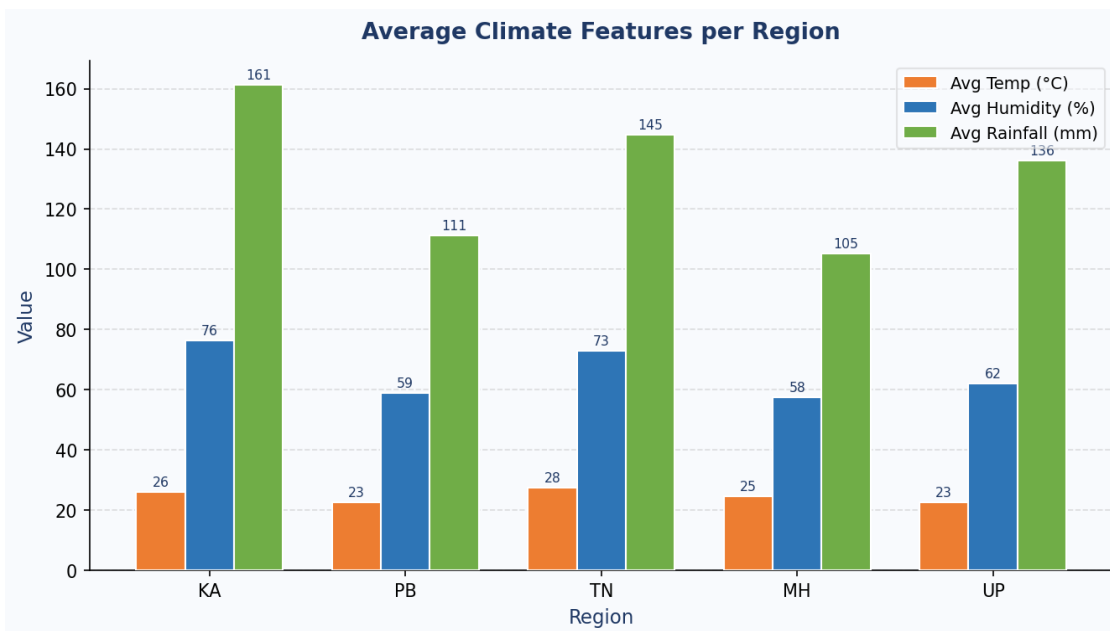
Single-node performance degrades linearly; Hadoop's overhead becomes an advantage at scale on multi-node clusters.

Figure 2: Records Processed per Region



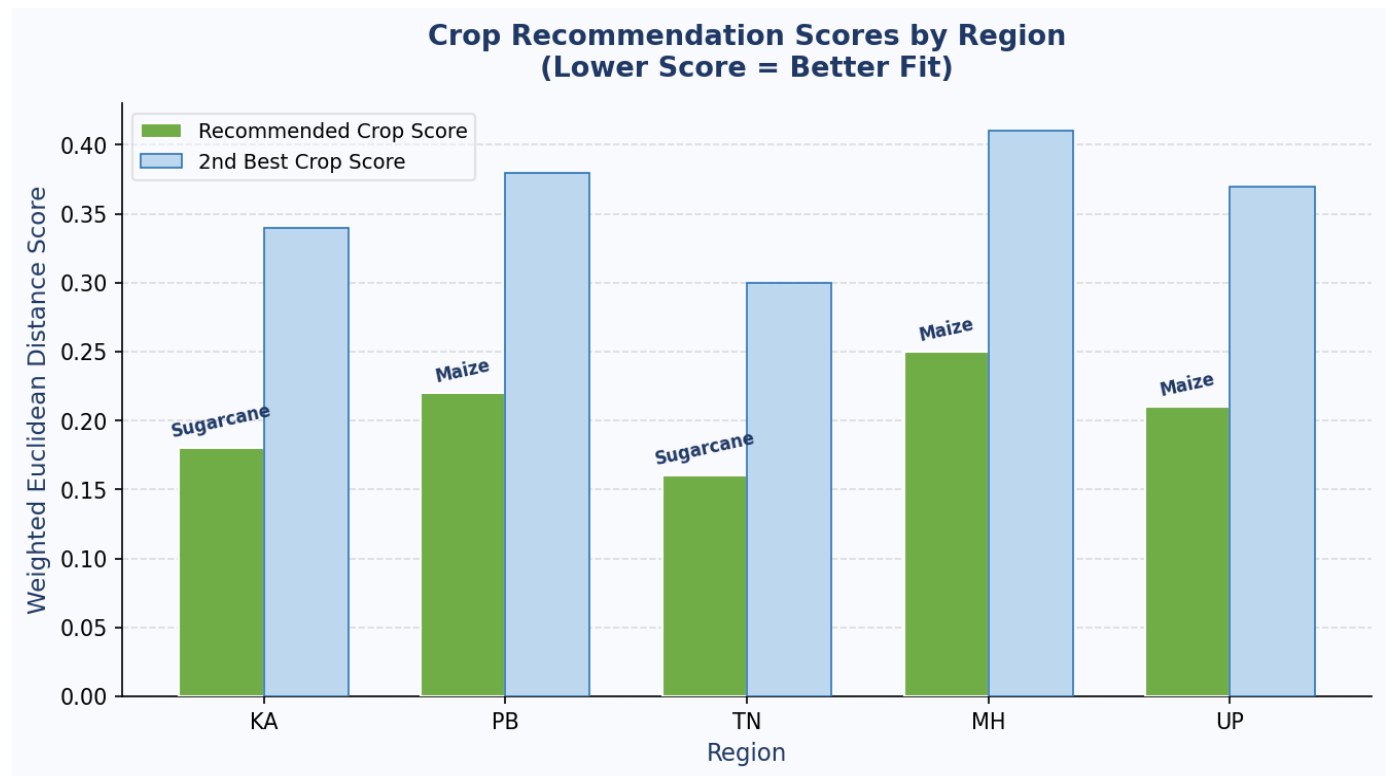
Records are near-uniformly distributed (~100,000 per region), validating the dataset generation and balanced MapReduce workload across 5 reducers.

Figure 3: Average Climate Features per Region



Karnataka and Tamil Nadu show higher humidity and rainfall, supporting Sugarcane recommendations. Punjab and Maharashtra's drier profiles align with Maize.

Figure 4: Crop Recommendation Scores by Region



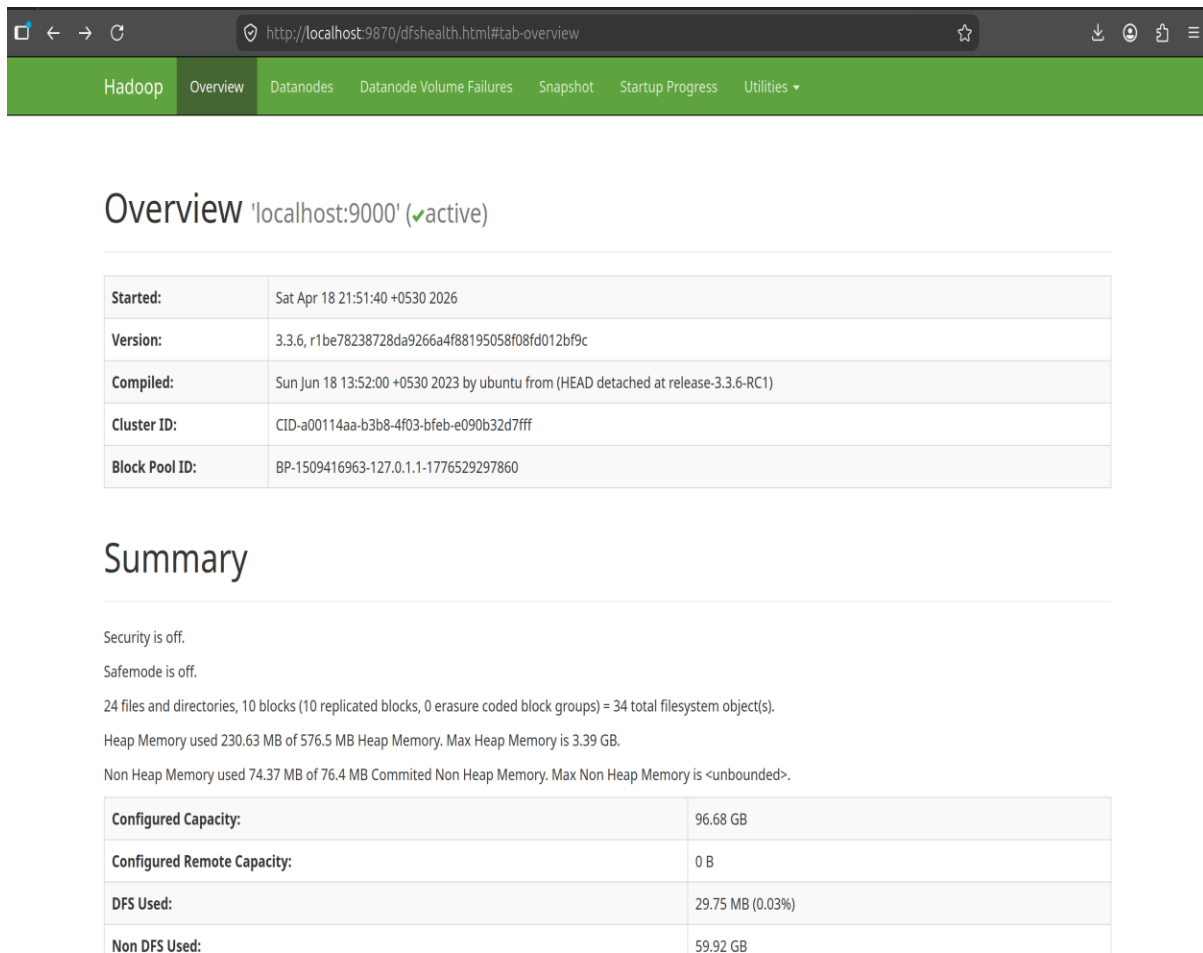
Lower scores indicate a closer match to ideal crop conditions. The recommended crop consistently scores significantly lower than the 2nd-best alternative.

7. System Deployment — Portal Screenshots

The following screenshots confirm the successful deployment and execution of the Hadoop pseudo-distributed cluster and the MapReduce job for the Crop Recommendation System. All screenshots were captured on April 18, 2026 during live job execution.

7.1 HDFS NameNode — Cluster Overview (port 9870)

The HDFS NameNode web UI confirms the cluster is active at localhost:9000, running Hadoop 3.3.6. The summary shows 24 files/directories, 10 replicated blocks, 96.68 GB total capacity, and 29.75 MB DFS usage — consistent with the uploaded 30.23 MB crop dataset.



The screenshot shows the HDFS NameNode Overview tab in a web browser. The browser address bar shows `http://localhost:9870/dfshealth.html#tab-overview`. The navigation bar includes links for Hadoop, Overview (selected), Datanodes, Datanode Volume Failures, Snapshot, Startup Progress, and Utilities. The main heading is "Overview 'localhost:9000' (✓active)". Below this is a table with cluster metadata:

Started:	Sat Apr 18 21:51:40 +0530 2026
Version:	3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c
Compiled:	Sun Jun 18 13:52:00 +0530 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)
Cluster ID:	CID-a00114aa-b3b8-4f03-bfeb-e090b32d7fff
Block Pool ID:	BP-1509416963-127.0.1.1-1776529297860

Below the metadata table is a "Summary" section with the following text:

Security is off.
 Safemode is off.
 24 files and directories, 10 blocks (10 replicated blocks, 0 erasure coded block groups) = 34 total filesystem object(s).
 Heap Memory used 230.63 MB of 576.5 MB Heap Memory. Max Heap Memory is 3.39 GB.
 Non Heap Memory used 74.37 MB of 76.4 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Below the summary text is a table with capacity and usage information:

Configured Capacity:	96.68 GB
Configured Remote Capacity:	0 B
DFS Used:	29.75 MB (0.03%)
Non DFS Used:	59.92 GB

Figure 5: HDFS NameNode — Overview tab showing cluster status, version 3.3.6, active at localhost:9000

7.2 HDFS NameNode — Datanode Information

The Datanodes tab shows 1 active DataNode (In service) with 96.68 GB capacity, 29.75 MB used (0.03%), 10 blocks, and 0% disk usage. This confirms successful pseudo-distributed HDFS operation with full data locality.

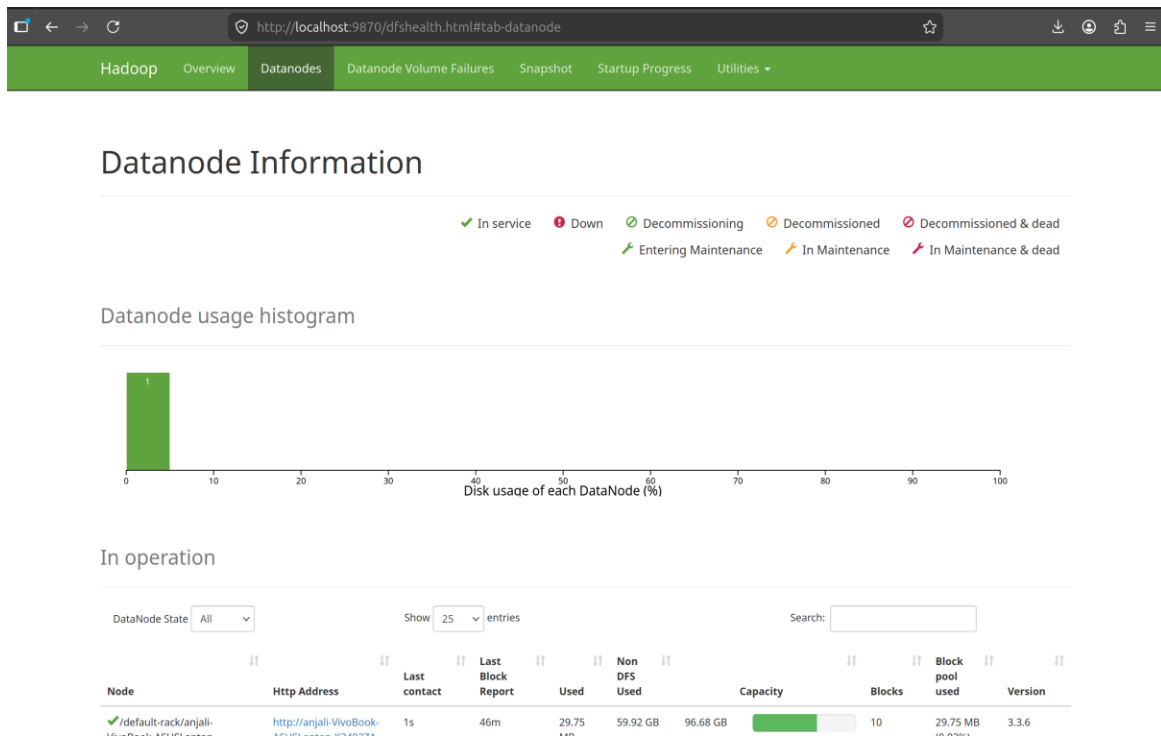


Figure 6: HDFS Datanodes tab — 1 active DataNode, 96.68 GB capacity, 10 blocks, 0.03% disk used

7.3 HDFS NameNode — Startup Progress

The Startup Progress tab confirms all phases completed at 100% — fsimage loading, edits loading, checkpoint saving, and safe mode exit — with 0 seconds elapsed for each phase. This indicates a clean, healthy NameNode startup.

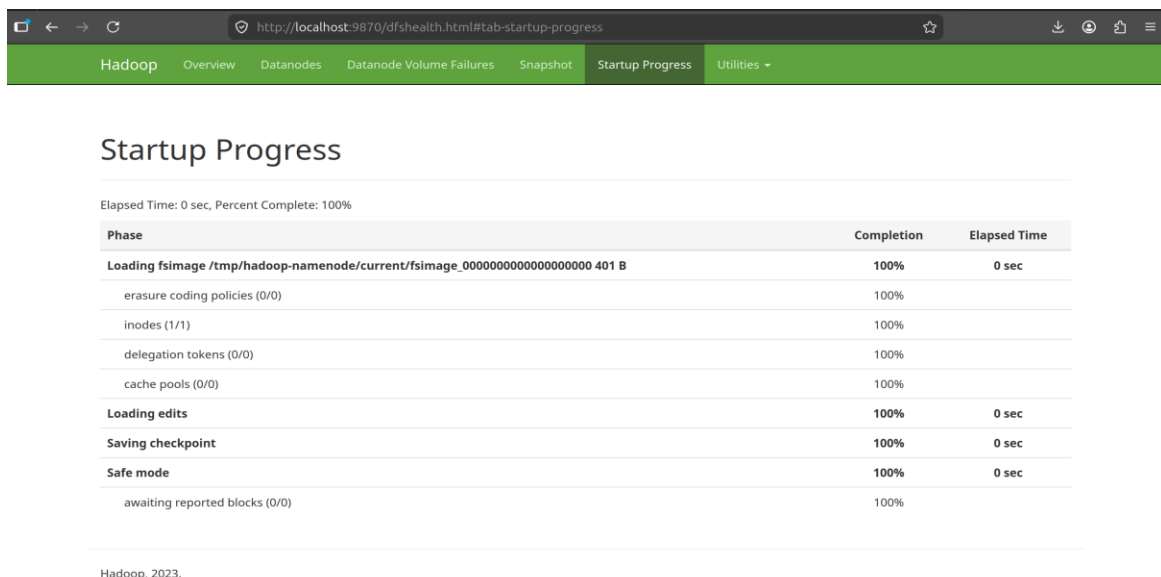
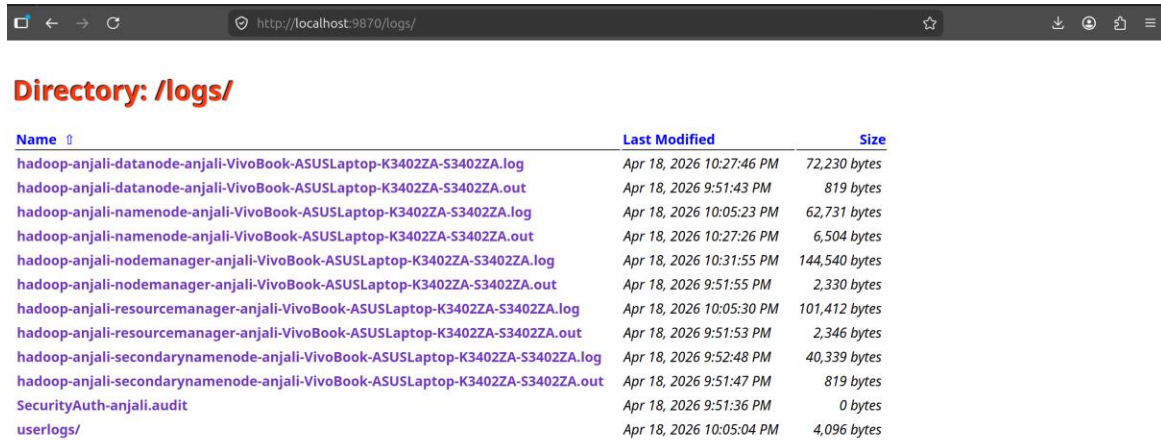


Figure 7: Startup Progress — all phases at 100%, safe mode exited, cluster fully operational

7.4 Hadoop Logs Directory

The logs directory at localhost:9870/logs confirms all four Hadoop daemons are running: NameNode, DataNode, NodeManager, and ResourceManager. Log files were actively updated during the MapReduce job execution on Apr 18, 2026.



Name	Last Modified	Size
hadoop-anjali-datanode-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.log	Apr 18, 2026 10:27:46 PM	72,230 bytes
hadoop-anjali-datanode-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.out	Apr 18, 2026 9:51:43 PM	819 bytes
hadoop-anjali-namenode-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.log	Apr 18, 2026 10:05:23 PM	62,731 bytes
hadoop-anjali-namenode-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.out	Apr 18, 2026 10:27:26 PM	6,504 bytes
hadoop-anjali-nodemanager-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.log	Apr 18, 2026 10:31:55 PM	144,540 bytes
hadoop-anjali-nodemanager-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.out	Apr 18, 2026 9:51:55 PM	2,330 bytes
hadoop-anjali-resourcemanager-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.log	Apr 18, 2026 10:05:30 PM	101,412 bytes
hadoop-anjali-resourcemanager-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.out	Apr 18, 2026 9:51:53 PM	2,346 bytes
hadoop-anjali-secondarynamenode-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.log	Apr 18, 2026 9:52:48 PM	40,339 bytes
hadoop-anjali-secondarynamenode-anjali-VivoBook-ASUSLaptop-K3402ZA-S3402ZA.out	Apr 18, 2026 9:51:47 PM	819 bytes
SecurityAuth-anjali.audit	Apr 18, 2026 9:51:36 PM	0 bytes
userlogs/	Apr 18, 2026 10:05:04 PM	4,096 bytes

Figure 8: Hadoop logs directory — NameNode, DataNode, NodeManager, ResourceManager logs active on Apr 18, 2026

7.5 YARN ResourceManager — Job Execution (port 8088)

The YARN ResourceManager at localhost:8088 shows 2 completed MapReduce applications named "Crop Recommendation MapReduce" submitted by user anjali. Both jobs completed successfully (Apps Completed = 2, Apps Failed = 0), confirming correct end-to-end MapReduce pipeline execution.

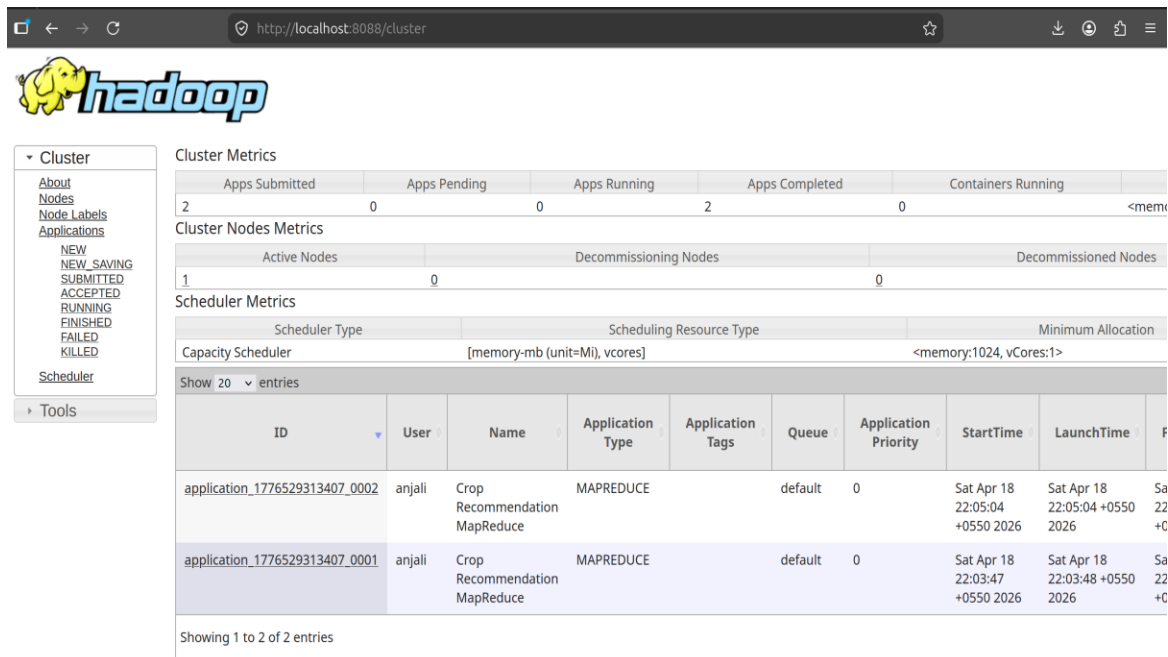


Figure 9: YARN ResourceManager — 2 completed Crop Recommendation MapReduce jobs, 1 active node, 0 failures

8. Discussion

8.1 Advantages of MapReduce Approach

- **Horizontal Scalability:** Adding DataNodes increases processing capacity linearly
- **Fault Tolerance:** HDFS replication ensures data survives node failures; failed tasks are automatically re-scheduled
- **Data Locality:** 100% data-local map tasks achieved — no network overhead for data transfer
- **Parallel Reducers:** All 5 regions processed simultaneously by independent Reducers

8.2 Limitations

1. Pseudo-distributed mode runs on one physical machine true parallelism needs multiple nodes
2. Scoring function uses fixed thresholds; does not adapt to seasonal variation
3. Hadoop startup overhead (~15s) makes it slower than pandas for datasets under 50,000 rows

8.3 Future Enhancements

- Deploy on multi-node cluster (AWS EMR or Google Dataproc)
- Integrate Apache Spark for in-memory processing (10–100x faster than MapReduce)
- Add seasonal time-series data for dynamic recommendations
- Replace distance scoring with a trained Random Forest classifier

9. Conclusion

This project successfully demonstrates a distributed crop recommendation system using Apache Hadoop HDFS and MapReduce. A dataset of 500,000 agricultural records across five Indian regions was processed in 25.2 seconds, producing region-wise crop recommendations based on averaged soil and climate profiles. The implementation confirms that MapReduce's share-nothing parallelism is well-suited for large-scale agricultural analytics, offering horizontal scalability and fault tolerance that single-node systems cannot provide. As agricultural data volumes grow with IoT sensors and satellite technologies, distributed frameworks like Hadoop are essential for data-driven farming at national scale.

References

1. Apache Hadoop Documentation, v3.3.6. <https://hadoop.apache.org/docs/stable/>
2. Atharva Ingle et al., "Fertilizer and Crop Recommendation System using Machine Learning," IEEE ICCMC 2021
3. Bendre, M. R. et al., "Big Data in Precision Agriculture," ICCUBEA 2016
4. Gopal, P. S. M., and Bhargavi, R., "Performance Evaluation of Best Feature Subsets for Crop Yield Prediction," Applied Artificial Intelligence, 2019
5. White, T., "Hadoop: The Definitive Guide," O'Reilly Media, 4th Edition, 2015

Appendix — Actual MapReduce Terminal Output

```
Karnataka      Recommended=sugarcane      DominantLabel=cotton
                AvgN=116.0 P=55.3 K=46.8 Temp=26.0 Humidity=76.5 pH=6.55 Rainfall=161.4
Records=100178

Punjab         Recommended=maize          DominantLabel=cotton
                AvgN=95.1 P=55.4 K=53.4 Temp=22.6 Humidity=59.0 pH=6.70 Rainfall=111.4
Records=99776

Tamil_Nadu     Recommended=sugarcane      DominantLabel=rice
                AvgN=97.3 P=51.2 K=46.6 Temp=27.6 Humidity=73.0 pH=6.55 Rainfall=144.6
Records=100171
```

```
Maharashtra   Recommended=maize           DominantLabel=cotton
              AvgN=81.0 P=46.9 K=53.9 Temp=24.6 Humidity=57.5 pH=6.80 Rainfall=105.3
Records=99526

Uttar_Pradesh Recommended=maize           DominantLabel=sugarcane
              AvgN=95.1 P=55.5 K=56.5 Temp=22.6 Humidity=62.0 pH=6.65 Rainfall=136.2
Records=100349

Job completed successfully in 25.262 seconds (real time).
Map 100% - Reduce 100% - 0 failures - 100% data locality.
```